Year 10 Science (Adjusted)
Semester 1 Examination 2020 - Revision

Earth and Space Sciences

1. Complete the following table.

TERM	DEFINITION
light year	Distance light travels in one year.
constellation	Group of stars forming a pattern.
galaxy	Collection of stars held together by gravity.
nebula	Gas cloud.
supernova	Massive explosion of a massive star near the end of its life.
red giant	Medium stars swell in size as they run out of fuel.
black hole	Very dense and huge gravity. Light can't escape.
fusion	Process producing energy in stars - hydrogen becomes helium.

2. Light travels at ***300,000 km/second***. A star is 97.6 light years away. We can only travel at about ***15 - 20 km/second*** with our best rockets. No calculations are necessary.

   (a) With our current technology, are we likely to be able to travel to this star in the near future? Give a few reasons for your answer.

   YES / NO (Circle your answer.)

   Reasons:

   - Spaceship is far too slow.
   - The distance is huge and will take hundreds of years to reach.
   - Will die well before we get there.

(b) Suggest ***one way*** we might be able to overcome the time factor of travelling to other galaxies, given it could be hundreds of years.

- Develop craft that can travel near the speed of light.
- Freeze people and wake them up hundreds of years later.
- Learn how to warp space and fold it on itself - makes the distance much shorter.

3. (a) What is the major element found in the Sun? hydrogen

(b) What happens to this element so that the Sun produces its energy? (HINT: Think about the fusion process.)

Hydrogen atoms collide and become helium - needs 1 million °C to do this. Energy is released.

(c) The Sun loses 5 million tonnes of mass per second. According to Einstein's equation ($E = mc^2$), what does this mass become?

The lost mass becomes the energy that the Sun emits.

4. How is the surface temperature of a star related to its colour?

TEMPERATURE	COLOUR
very hot	blue
cool	red
moderate	yellow

5. (a) About how old is our Sun (in millions of years)? 5000 million years

(b) How much longer will it "live"? 5000 million years

6. Use the following words to complete the following flow diagrams.

white dwarf ***black dwarf*** ***nova*** ***supernova*** ***red giant***

black hole ***red supergiant*** ***neutron star***

(a) Draw a flow diagram (with arrows) to show the stages of our Sun's remaining life as it runs out of fuel.

Sun → red giant → nova → white dwarf → black dwarf

(b) Do similar flow diagrams for the following types of stars.

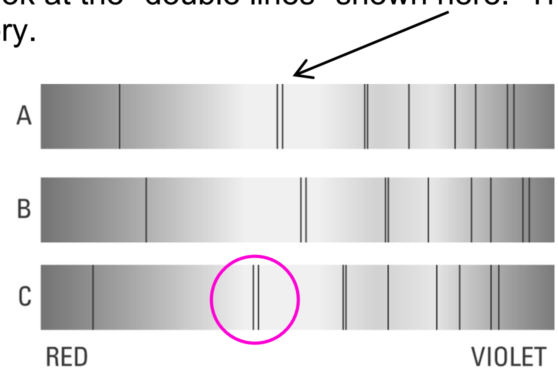
Huge star → red supergiant → supernova → neutron star

Giant star → red supergiant → supernova → black hole

7. Give two characteristics of a ***black hole***.

- Very large gravity.
- Light cannot escape from it.

8. The diagram below shows the spectra (black lines) seen on the light coming from distant galaxies. Look at the "double lines" shown here. This shows the lines as seen in our laboratory.



Look at B and C above and where the double lines have moved.

(a) Circle the lines that have been "red-shifted".

(b) Does "red-shift" mean the galaxy is moving towards us? Circle your answer.

YES NO

9. Below is a list of the four spheres on Earth. Give a brief description of each. i.e. What do you find in each one?

Biosphere: Made up of all living things - plant and animals.

Atmosphere: Made up of the troposphere (lower atmosphere) and stratosphere (upper atmosphere). Contains the ozone layer that protects the planet from UV rays.

Hydrosphere: Made up of all the water on Earth, including that frozen in ice, liquid as water in oceans and contained in the atmosphere.

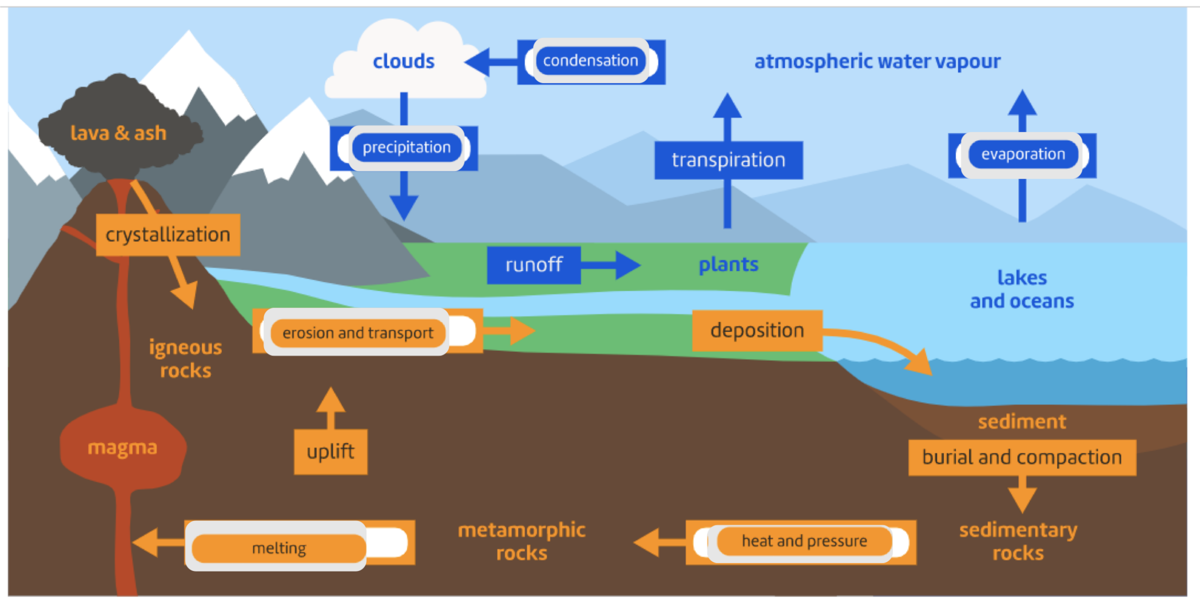
Lithosphere: Made up of the rocky crust and soil on the Earth.

10. Describe two consequences of ***global warming***.

- Rising sea levels. Thawing of permafrost (frozen ground). Species extinction due to habitat change.
- Melting ice caps. Melting glaciers. More droughts, fires, etc.

11. The following diagram shows the rock cycle and the water cycle. Complete the diagram with appropriate labels.

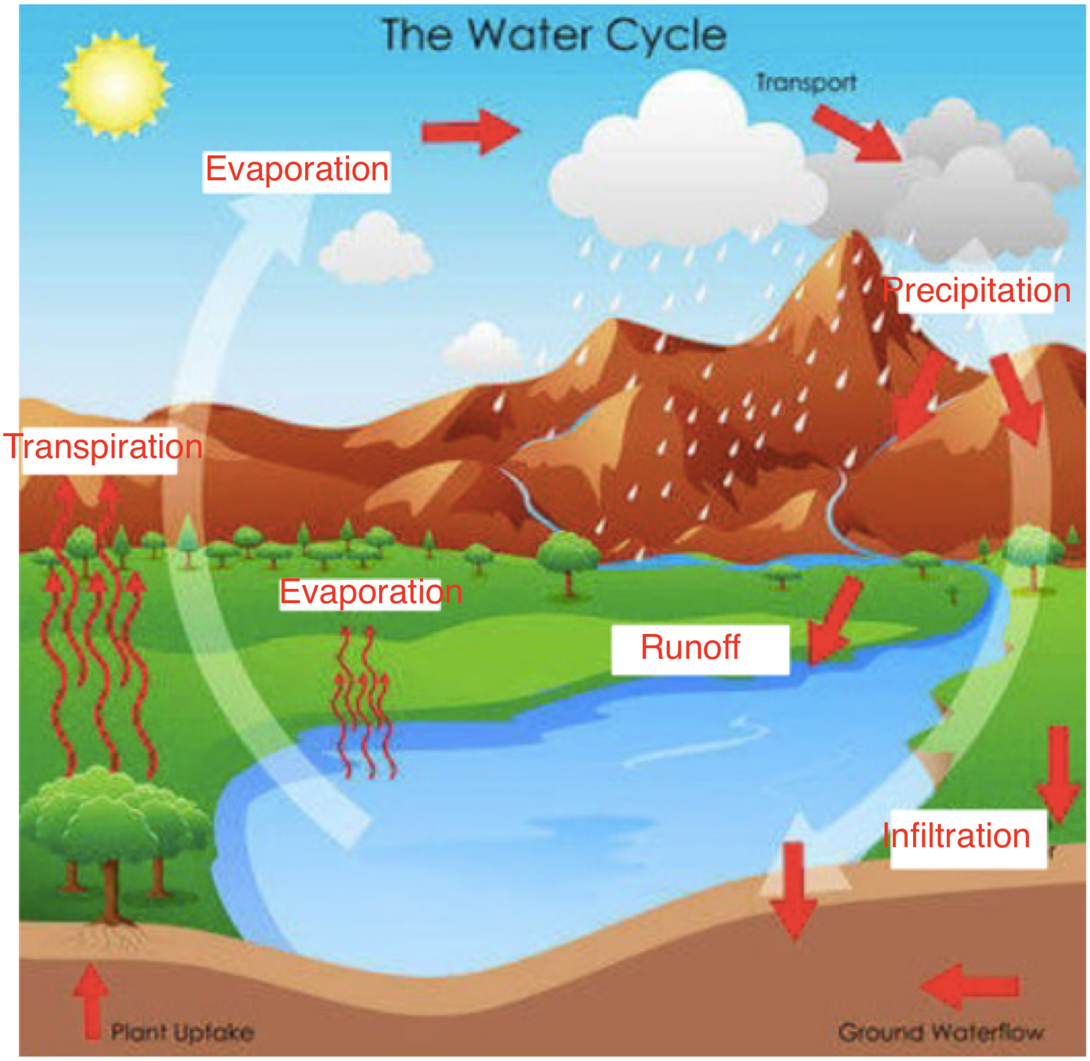
evaporation | precipitation | condensation | melting | heat and pressure | erosion and transport



12. (a) Label the diagram below showing the water cycle. Us the following words.

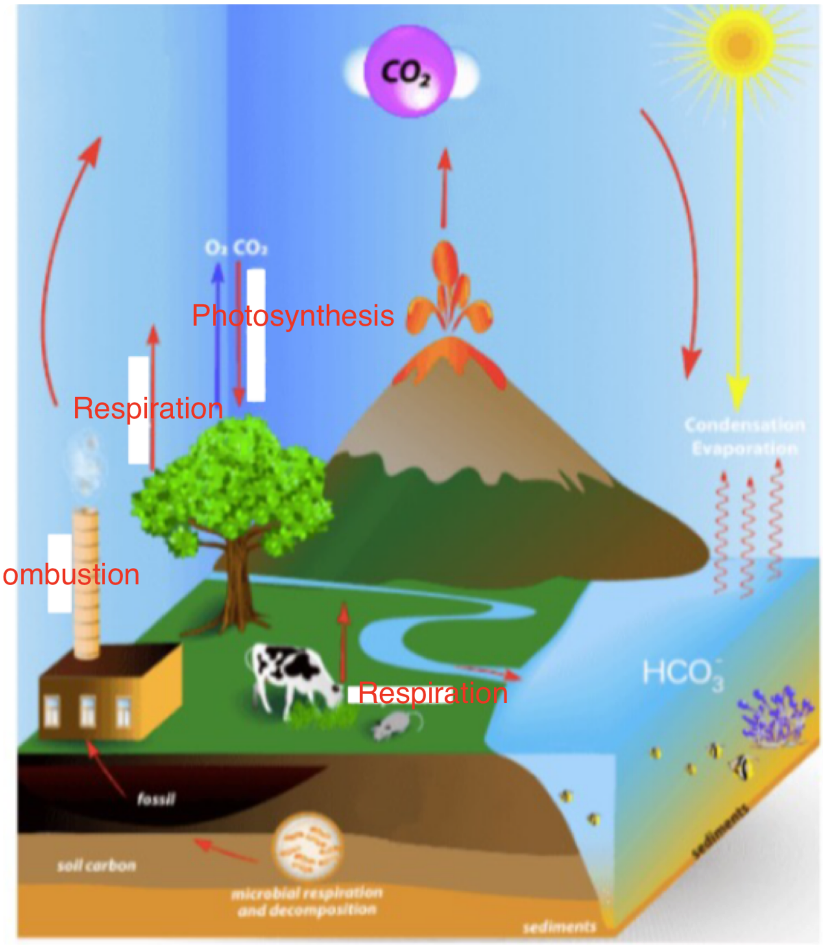
evaporation ***infiltration*** ***precipitation*** ***runoff***

transpiration ***condensation***



13. The following diagram shows the carbon cycle. Label the processes missing from the diagram using the words below.

combustion ***photosynthesis*** ***respiration***



14. Use the words in the following to complete the sentences.

water rocks lithosphere glaciers hydrosphere

The Earth' hydrosphere ; made up of all the water on our planet. This includes all of the salt water oceans, fresh water lakes, glaciers and underground deposits.

The lithosphere ncludes all rocks and geological structures on which life exists, such as soil, rocks and mountains.

gases sun biosphere stable atmosphere unstable weather

The biosphere is made up of all areas of the earth and its atmosphere that contain life.

The atmosphere contains all the gases above Earth. Parts of the atmosphere contain the weather and greenhouse gases that keep Earth's temperatures stable.

ocean loses evaporates gains condenses

Water in the ocean is heated by the sun. It evaporates and forms water vapour.

Clouds form when water vapour condenses. This happens because the water vapour loses heat energy as it rises.

quickly ocean evaporate rain puddle

When water precipitates as rain it either soaks into the ground or falls into a lake, river or ocean. If it falls into an ocean it could take a very long time to evaporate but if it falls into a puddle it could evaporate very quickly.

flows Runoff Infiltration

Infiltration occurs when water is absorbed by the soil.

Runoff occurs when water flows across the ground into a lake, river, or ocean.

leaves | transpiration | flowers | water vapour | steam | infiltration | roots

Water is returned to the water cycle after it is absorbed by soil into the roots of a plant, it then travels through the plant and escapes through the leaves of the plant as water vapour

This is called transpiration

Human Biology

1. (a) What are the male and female ***gametes*** in animals called?

sperm and egg (ovum)

(b) What are the ***genotypes*** for a male and a female?

male XY female XX

2. DNA is thought to be a twisted ladder structure called a double helix. There are a series of bases located between the two strands of the helix.

(a) Name the four bases.

guanine (G) thymine (T) adenine (A) guanine (G)

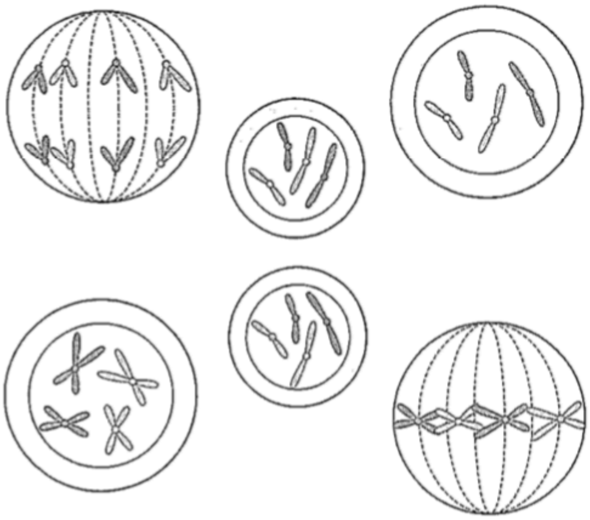
(b) Which bases are paired together?

A - T
G -C

3. Define the following terms.

allele	Alternative forms of a gene. e.g. T, t
pure-bred	2 alleles for a gene are the same. e.g. TT, tt
hybrid	2 different alleles. e.g. Tt
homozygous	Pure-bred - alleles are the same. e.g. TT, tt
heterozygous	Hybrid - alleles are different. e.g. Tt
genotype	The genes for a characteristic. e.g. TT, Tt, tt
phenotype	How the organism looks. e.g. Tall, dwarf
sex-linked	Characteristic carried on the X or Y chromosome.
dominant	A gene that "masks" another. e.g. T dominates t
recessive	A gene that doesn't show if the dominant gene is present.
co-dominant	Both characteristics show. e.g. Red + white hair in cattle.
incomplete dominance	New characteristic shows - a "blend". e.g. Pink colour when red + white flowers are crossed.
haploid	Half the normal number of chromosomes.
gene	Information for a specific characteristic.

4. Below is a diagram showing the ***mitosis*** cell division process. The stages are in the wrong order. Try to put them in the correct order in the blank diagram.



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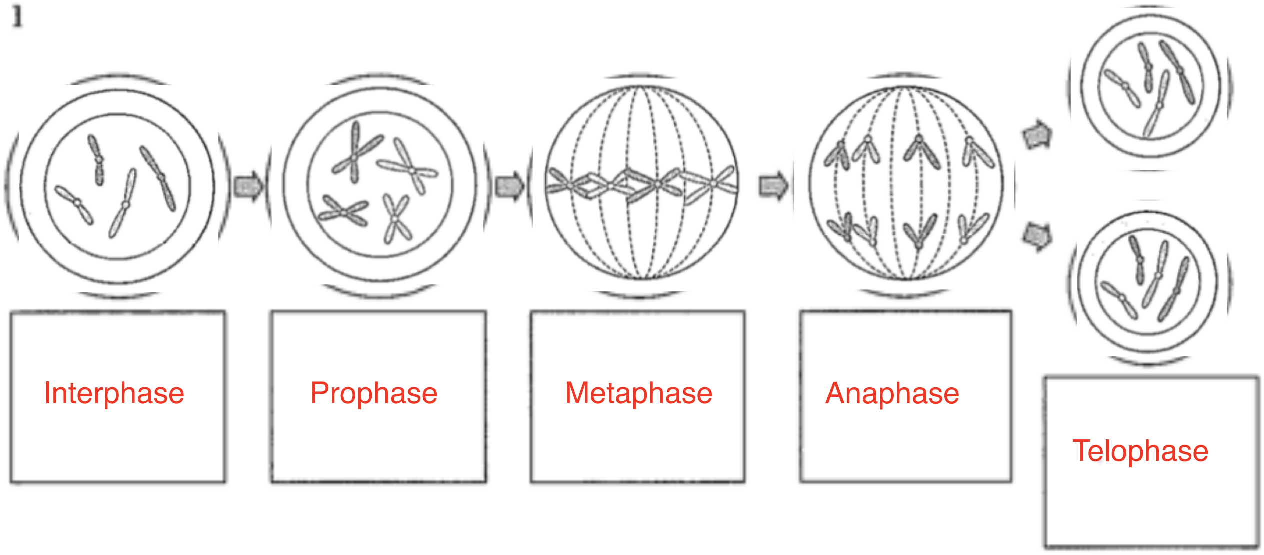


5. Polled cattle do not have horns. This is a ***dominant*** phenotype arising from the action of a gene **P**. Horned cattle are therefore **pp** hybrids.

What are the offspring of a hybrid polled bull (Pp) crossed with a pure-bred horned cow (pp)? ***Show full working***.

Pp × pp
♂ ♀

♀ \ ♂	P	p
p	Pp	pp
p	Pp	pp

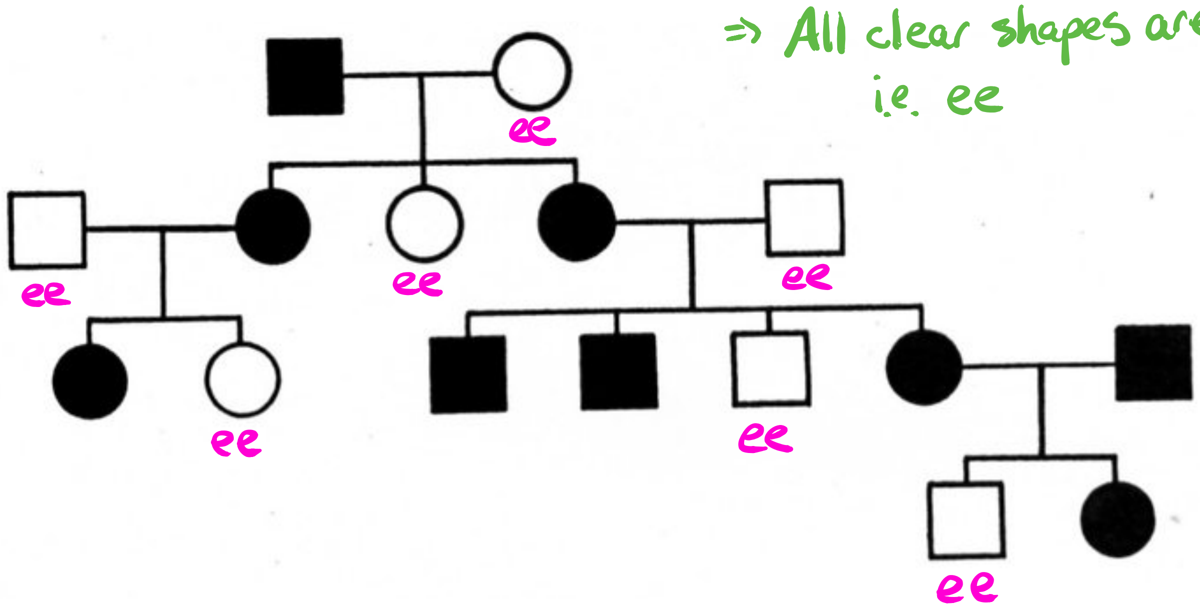
$\underbrace{\text{Pp Pp}}_{\text{50\% polled}}$ $\underbrace{\text{pp pp}}_{\text{50\% horned}}$

6. The following pedigree shows the occurrence of no ear lobes in a family. It is a ***dominant*** trait.

i.e. E = no ear lobes e = ear lobes

Dominant trait

⇒ All clear shapes are recessive i.e. ee



(a) How many males have no ear lobes? 4 (squares)

(b) How many females have lobes? 5 (circles)

(c) What is the genotype for an individual with lobes? ee

7. (a) How many chromosomes are found in a human skin cell (a normal cell)?

46

(b) How many chromosomes are found in the reproductive cells?

23

(c) Name the process used to make reproductive cells? meiosis

8. (a) Who determines the sex of a child - the mother or father? Explain why.

Father - he has the Y chromosome that determines maleness.

(b) A family has had five boys born into it, and the mother is pregnant with a sixth child. What is the chance that the child will be a girl?

50:50 (50% chance)